

Documentation (Roman Urdu)

Yeh AI engine ka "Zero-Latency Router" ya smart classifier hai. Project mein performance barqarar rakhne ke liye har choti bat ko LLM (Artificial Intelligence) se classify nai krvaaya ja sakta, werna bot slow hojye ga. Yeh pure regex algorithms aur pattern matching se user ki bat (intent) samajhta hai.

File Ka Maqsad (Purpose)

Har message par system yeh soachta hai ke usay purani chat history bhejni chahiye ya nahi? Aur kia isay documents search (RAG) krney hein? Iska faisla yeh script karti hai.

Code Blocks aur Functions ki Tafseel

1. Regex Pattern Arrays

File ke upr mukhtalif list banayi gai hain: * `FULL_HISTORY_PATTERNS`: Wo alfaaz jisme user puri chat k hawala deta ha (jaise "remember earlier", "summary of our chat"). * `RAG_PATTERNS`: Aisy alfaaz jinme RAG zaroori parta ha (jaise "in the document", "search the pdf"). * `STANDALONE_PATTERNS`: Aisi batian jisme context/history ki zrurat nai hoti (Greetings like "hello", "thanks", ya standalone functions "write a poem"). * `RECENT_CONTEXT_INDICATORS`: Continuity wale lafz (jaise "yes", "and also", "fix this").

2. `def classify(message, history_length)`

- **Kya karta hai?:** Jab koi request ati hai toh wo list se matching function call krti ha.

Logic (Faislay ki tartib):

1. Sub se pehlay **RAG** check krta ha. Agr match mily tou (`history_depth=4`, `needs_rag=True`) de kar bata deta hai k docs padhney hein.
2. Phr **FULL HISTORY** pattern dekhta ha. Agr result han m ay to `history_depth=-1` mark karta hai jiska mtlb ha sari available messages ki line history mein bhijwao LLM ko.
3. Phir standalone check karta ha k purani batein waste ho jayen prompt limit save krny k liy.
4. Akhir me default behavior de kr recent history set kr deta ha. Resulting dictionary se main logic app samajh bna leti ha.

